Assignment

Print the Equilibrium Index for the input:

```
n = 5
arr = {2, -4, -2, 4, 2}
```

```
 Rikhil Nellimarla on TARS at …\glitch-clicker via  main via J v24.0.2 via  v3.13.7
 java Main
Enter array size: 5
Enter 5 elements: 2 -4 -2 4 2

--- Result ---
Maximum Equilibrium Sum: 2
Equilibrium Index: 0
```

Code

```java
import java.util.*;

class Main{

    public static void main(String args[])
    {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter array size: ");
        int n = sc.nextInt();
        int array[] = new int[n];
        int totalSum = 0;

        System.out.print("Enter " + n + " elements: ");
        for(int i = 0; i < n; i++)
        {
            array[i] = sc.nextInt();
            totalSum += array[i];
        }

        int maxEquilibriumSum = Integer.MIN_VALUE;
        int equilibriumIndex = -1;
        int prefixSum = 0;
        int suffixSum = totalSum;

        for(int i = 0; i < n; i++)
        {
            prefixSum += array[i];
            if(prefixSum == suffixSum)
            {
```

```java
            if(prefixSum > maxEquilibriumSum)
            {
                maxEquilibriumSum = prefixSum;
                equilibriumIndex = i;
            }
        }
        suffixSum -= array[i];
    }

    sc.close();

    System.out.println("\n--- Result ---");

    if(equilibriumIndex != -1)
    {
        System.out.println("Maximum Equilibrium Sum: " + maxEquilibriumSum);
        System.out.println("Equilibrium Index: " + equilibriumIndex);
    }
    else
    {
        System.out.println("No equilibrium point found");
    }
  }
}
```

Time and Space Complexities

Time Complexity: O(n)
Space Complexity: O(n)